

knowing the latest vscode version number, i'd like a way to grab the corresponding tag's commit sha on github. how do i crawl through github's API to do that?

Nachgedacht für 23s

Got it. You already know the VS Code version string (e.g., 1.94.2) and you want the **commit SHA** of the corresponding Git tag in microsoft/vscode .

The GitHub REST API makes this straightforward. The key is: a tag reference can point either directly to a **commit** (lightweight tag) or to a **tag object** (annotated tag) that then points to the commit. So you need to handle both.

Below are copy-pasteable solutions in Bash/curl+jq and PowerShell.

Bash (curl + jq)

bash

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```
# Inputs
OWNER="microsoft"
REPO="vscode"
VERSION="1.94.2"      # your version
GITHUB_TOKEN=""       # optional but recommended to avoid rate limits

authHeader=()
[ -n "$GITHUB_TOKEN" ] && authHeader=(-H "Authorization: Bearer $GITHUB_TOKEN")

# 1) Try an exact ref match
refJson=$(curl -s "${authHeader[@]}" \
  "https://api.github.com/repos/$OWNER/$REPO/git/ref/tags/$VERSION")

# If that 404s (e.g. tag prefix differences), fall back to matching-refs
if echo "$refJson" | jq -e 'has("object")' >/dev/null 2>&1; then
:
else
  refJson=$(curl -s "${authHeader[@]}" \
```

```

"https://api.github.com/repos/$OWNER/$REPO/git/matching-
refs/tags/$VERSION")

# pick the first exact or best match
# (tighten the filter if your tag naming scheme is stricter, e.g.
select(.ref=="refs/tags/1.94.2"))
refJson=$(echo "$refJson" | jq -e '.[0]') || {
    echo "Tag not found for '$VERSION'; exit 1;
}
fi

type=$(echo "$refJson" | jq -r '.object.type')
sha=$(echo "$refJson" | jq -r '.object.sha')

if [ "$type" = "commit" ]; then
    # Lightweight tag: done
    COMMIT_SHA="$sha"
else
    # Annotated tag: dereference the tag object to reach the commit
    tagObj=$(curl -s "${authHeader[@]}" \
        "https://api.github.com/repos/$OWNER/$REPO/git/tags/$sha")
    COMMIT_SHA=$(echo "$tagObj" | jq -r '.object.sha')
fi

echo "$COMMIT_SHA"

```

Notes

- If VS Code's tags use a `v` prefix in some repos, set `VERSION="v1.94.2"` instead. If unsure, the `matching-refs` fallback handles it.
- Auth is optional, but passing a token avoids low rate limits.

PowerShell (Invoke-RestMethod)

powershell

 Copy code

```

$Owner = "microsoft"
$Repo = "vscode"
$Version = "1.94.2" # your version string
$Token = $env:GITHUB_TOKEN # optional

$headers = @{}

```

```

if ($Token) { $Headers["Authorization"] = "Bearer $Token" }

function Get-Json($Url) {
    try { Invoke-RestMethod -Headers $Headers -Uri $Url -Method GET -
ErrorAction Stop }
    catch { $null }
}

# 1) Try exact ref
$ref = Get-Json
"https://api.github.com/repos/$Owner/$Repo/git/ref/tags/$Version"

# 2) Fallback: matching-refs if exact not found
if (-not $ref) {
    $refs = Get-Json "https://api.github.com/repos/$Owner/$Repo/git/matching-
refs/tags/$Version"
    if (-not $refs) { throw "Tag not found for '$Version'" }
    # choose the first; refine selection if you need exact equality:
    # $ref = $refs | Where-Object { $_.ref -eq "refs/tags/$Version" } |
Select-Object -First 1
    $ref = $refs | Select-Object -First 1
}

switch ($ref.object.type) {
    "commit" { $commitSha = $ref.object.sha }
    "tag" {
        $tagObj = Get-Json
"https://api.github.com/repos/$Owner/$Repo/git/tags/$($ref.object.sha)"
        if (-not $tagObj) { throw "Failed to dereference annotated tag." }
        $commitSha = $tagObj.object.sha
    }
    default { throw "Unexpected ref object type: $($ref.object.type)" }
}

$commitSha

```

Skip to content

One-liner variants (if you're sure about lightweight vs annotated)

- **Lightweight tag only** (fails on annotated tags):

bash

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```
curl -s -H "Authorization: Skip to content JKEN" \
  "https://api.github.com/repos/microsoft/vscode/git/ref/tags/1.94.2" \
  | jq -r '.object.sha'
```

- **Robust single line (jq handles both):**

bash

 Copy code

```
curl -s -H "Authorization: Bearer $GITHUB_TOKEN" \
  "https://api.github.com/repos/microsoft/vscode/git/ref/tags/1.94.2" \
  | jq -r 'if .object.type=="commit" then .object.sha else (.object.sha |
  tostring) end' \
  | xargs -I{} curl -s -H "Authorization: Bearer $GITHUB_TOKEN" \
    "https://api.github.com/repos/microsoft/vscode/git/tags/{}" \
  | jq -r '.object.sha? // empty'
```

(This prints the commit SHA for annotated tags; for lightweight tags the first jq branch already returns the commit SHA—adjust if you want to always print exactly once.)

Tips / gotchas

- **Tag naming:** Some projects use `v1.2.3` ; others use `1.2.3` . If you get 404s on the exact ref, query `git/matching-refs` as shown.
- **Rate limits:** Unauthed calls are tiny. Use a token via `Authorization: Bearer <token>` .
- **Releases vs tags:** If you *didn't* know the version, `GET /repos/:owner/:repo/releases/latest` gives you `tag_name` , which you can then resolve with the same steps.

If you want, I can wrap this into a tiny script/CLI that takes `owner/repo` and `version` and prints the commit SHA.



(using powershell) do i really need a github token?

Nope — a **GitHub token isn't required** if you only need occasional lookups.

But here's the catch:

- **Without a token** you get the *unauthenticated* quota: → **60 requests per hour, per IP**.
- **With a token** (any personal access token, even with no scopes set) you get the *authenticated* quota: → **5,000 requests per hour, per user**.

For your case — “I know the VS Code version, give me the commit SHA” — you’ll probably make **1–2 requests** per version. That’s totally fine without a token.

Only if you’re scripting this against many repos/versions, or running in CI/CD, would I recommend setting up a token.

Example: PowerShell without a token

powershell

 Copy code

```
$Owner = "microsoft"
$Repo = "vscode"
$Version = "1.94.2"

# No headers, just straight call
$ref = Invoke-RestMethod -Uri
"https://api.github.com/repos/$Owner/$Repo/git/ref/tags/$Version"

switch ($ref.object.type) {
    "commit" { $commitSha = $ref.object.sha }
    "tag" {
        $tagObj = Invoke-RestMethod -Uri
        "https://api.github.com/repos/$Owner/$Repo/git/tags/$($ref.object.sha)"
        $commitSha = $tagObj.object.sha
    }
}

$commitSha
```

👉 So unless you hit rate limits, you can skip the token entirely.

Do you want me to show you a minimal one-liner in PowerShell that works for both lightweight *and* annotated tags without bothering with token setup?



+ Ask anything



ChatGPT can make mistakes. Check important info.